



Karl-Franzens-Universität Graz

Business Plan

AI-Powered Clinical Decision Support System (CDSS)

An Analysis of Economic, Regulatory and Ethical Aspects of AI-Assisted
Medical Decision Supportcision Support.

Course

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Lecturer

Dipl. -Ing. (FH) MA, Michael Freidl

Prepared by

Julian Baron, Zan Porenta, Satera Sleinyte

Project Veritatem

Explainable AI for early-pregnancy gestational-diabetes risk

A business plan for an AI-based clinical decision-support product

Economic Aspects in AI — Seminar Paper

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Executive Summary

Veritatem is a proposed software-as-a-medical-device (SaMD) product that estimates a pregnant woman's risk of gestational diabetes mellitus (GDM) in the first trimester, using data that is already collected at the first antenatal visit, and presents that estimate together with a plain explanation of why the score is what it is and what the clinician might do next.

The problem it addresses is structural. Across Europe, GDM is usually diagnosed by an oral glucose-tolerance test (OGTT) at 24-28 weeks of pregnancy. By then much of the metabolic and fetal trajectory is already set, and the window for the cheapest, least invasive interventions (diet, activity, closer monitoring) has narrowed. Roughly one in nine to one in six pregnancies is affected, depending on population and diagnostic criteria, so this is a high-volume condition, not a rare one.

The bet behind Veritatem is that moving risk stratification months earlier, while keeping the clinician firmly in charge and making every recommendation explainable and auditable, produces better outcomes at acceptable cost. A recent randomized trial of early diagnosis and treatment provides encouraging, though not yet settled, evidence that acting before 24-28 weeks can both improve outcomes and save money.

The recommended commercial path is providers first (maternity units and antenatal clinics) on clinical and public-benefit grounds, with payers second as the route to funding and scale once real-world evidence accumulates. The market is entered EU-first, with an Austria–Slovenia beachhead where antenatal care is structured and the regulatory regime is the one the product is designed for.

For this business plan to work, we must secure local, consented training and validation data (this is the single largest dependency and risk). It must achieve regulatory clearance as a CE-marked medical device under a high-risk AI regime and the surrounding care pathway must be able to absorb the additional patients it flags. The algorithm itself is not the moat - comparable models are published in the academic literature. The defensible position is the combination of clearance, local validation, the early-detection window, and explainability, all of which depend on data the venture must still earn access to.

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1. Business Description & History

Veritatem is an early-stage, pre-commercial AI driven project. Its purpose is to bring the detection of gestational-diabetes risk forward from the second trimester into the first, and to do so in a form that clinicians can trust, scrutinize, and act on. The product is delivered as software: a risk model that runs on data captured at routine first-trimester antenatal contact and returns a risk tier, a per-patient explanation, and a guideline-linked suggested next step. It is decision support, not a diagnosis, and the clinician remains the decision-maker.

1.1 Clinical problem and timing

Gestational diabetes is hyperglycaemia first recognised in pregnancy. It is common: a systematic review of European prevalence put the weighted figure at roughly eleven percent of pregnancies, with wide regional variation, and global estimates cluster around one in six when the inclusive IADPSG criteria are applied. Hyperglycaemia in pregnancy raises the risk of macrosomia, caesarean delivery, neonatal complications, pre-eclampsia, and it materially raises the mother's long-term risk of type 2 diabetes (studies have reported progression rates as high as roughly a third of women within a year postpartum and a majority over longer follow-up).

The structural weakness in current practice is timing. The diagnostic OGTT is normally performed at 24-28 weeks. Yet there is evidence that the metabolic divergence between women who develop GDM and those who do not is already detectable in early pregnancy, and that fetal growth can begin to accelerate before the standard diagnosis is even made. Diagnosing late means intervening late, when the cheapest levers have lost some of their power.

It is important to note that the evidence that earlier action improves outcomes, while growing, is not yet of the volume or consistency that would make it beyond dispute. The strongest recent signal is a single international randomized trial of early diagnosis and treatment, whose economic analysis suggested early treatment was both more effective and tended to be cost saving. That is promising but not the final word. Screening criteria and thresholds for GDM are contested across Europe, which complicates any product that must define what “risk” means. Veritatem is built on the premise that the early window is valuable - if large trials were to show that earlier information does not change outcomes, the core thesis would be wrong. We treat that as the central risk to revisit, not a settled fact.

1.2 Value proposition

Two things distinguish Veritatem from a generic predictive model. The first is the early-detection window itself: a tool designed to run on first-trimester inputs, rather than re-scoring patients at the point where they

would be tested anyway. The second is explainability built in as a clinical safety feature rather than bolted on. Every score is accompanied by the factors that drove it and a guideline-linked suggested action, and every interaction is logged. In a regulated, high-stakes clinical setting, that transparency is not cosmetic - it is what makes the tool adoptable, auditable, and defensible.

1.3 Customer segments

The primary customer is the healthcare provider: maternity units, antenatal clinics, and the obstetric and diabetology teams within them. They are sequenced first deliberately, on clinical and public-benefit grounds - the people who can act on an early flag are clinicians. Payers (public and private insurers and health systems) are a secondary segment: they stand to gain from fewer avoidable complications and a better long-term metabolic-risk pathway, but that benefit accrues over years and is the route to funding scale rather than the first point of sale. EHR vendors are a tertiary, licensing-oriented segment.

1.4 Profitability

Veritatem could be profitable because most of its costs are fixed and occur before launch: data access, model development, clinical validation, regulatory clearance and specialist staff. Once the model is approved and integrated, the cost of screening one additional patient is very low, because the AI works with simple structured clinical data. Therefore, profitability depends on scale: if the tool is used across many clinics and countries, the initial fixed costs can be spread over a large number of screenings, improving margins. However, scaling is not effortless, because medical AI still requires ongoing monitoring, compliance, revalidation and local adaptation when entering new markets.

1.5 Business Structure

Veritatem would operate as a limited-liability company registered in an EU/EEA member state. This structure is suitable because the product is intended for the European healthcare market and would need to comply with EU medical-device regulation, the AI Act and GDPR. An EU-based entity could act as the legal manufacturer of the CE-marked software medical device and take responsibility for conformity assessment, data protection, quality management and post-market monitoring.

The limited-liability structure also fits the risk profile of the project, as regulated medical AI requires significant investment, clinical partnerships and legal accountability. As the company expands, additional local arrangements may be needed for reimbursement, healthcare-system integration or partnerships in specific markets.

2. Product/Service Description

2.1 Product offering

At the first antenatal visit, a clinician (or the clinic's information system) supplies the routine information already gathered at that point: maternal age, body-mass index, relevant history such as a previous GDM pregnancy or family history of diabetes, and standard early-pregnancy laboratory values. Veritatem returns three things: a risk tier (for example low / moderate / high) with a calibrated probability and an uncertainty band (a per-patient explanation showing which factors pushed the estimate up or down) and a guideline-linked suggested next step, such as offering an early OGTT or a lifestyle referral. The clinician decides what to do and can override the suggestion, recording the reason. Every step is written to an audit log.

2.2 AI model and clinical interpretation

Veritatem would use a machine-learning model designed for structured clinical data, such as age, BMI, medical history and early pregnancy lab values. Since this is tabular data rather than images or text, models such as gradient-boosted decision trees are suitable because they are efficient, work well with mixed medical variables and can handle some missing data.

A key part of the product is clinical interpretation. The system would not only give a risk score, but also show which factors influenced that score. For example, a clinician could see that a high-risk result was mainly driven by previous gestational diabetes, BMI or fasting glucose. This makes the tool more useful in clinical practice, because doctors need to understand and justify the recommendation rather than blindly follow an AI result.

Once the model is built, each additional screening is very cheap to run because the calculation is simple. However, because the product would be used in a regulated medical setting, the model could not change automatically or retrain silently. Each version would need to be controlled, logged and updated through a formal process to ensure safety, traceability and compliance.

2.3 Key resources and competitive advantage

Veritatem depends on several key resources. The company would need clinical data scientists and ML engineers to build accurate, explainable models, as well as regulatory and clinical-affairs experts to guide the product through conformity assessment. Its technical resources would include the trained model, explanation layer, clinician dashboard and EHR integration code. These would be supported by intellectual property, internal know-how and an ISO 13485 quality-management system. However, these resources only

become valuable if the company can secure the most important resource: local, consented, high-quality training and validation data from EU (specifically Slovenian and Austrian) populations.

The algorithm itself is not the moat. Comparable GDM-prediction models, including explainable ones, are published and reproducible, so a competent team could rebuild the core model. What is genuinely hard to copy is the combination of:

- A CE-marked and AI-Act-compliant product
- Validation on local populations that regulators and clinicians in those markets will accept
- The early-detection positioning and the explainability and audit trail that make adoption defensible

Each piece is contingent on data and on regulatory investment, and it is the bundle, not any single component, that is defensible.

2.4 Competitor analysis

The most dangerous competitor is not another product but the status quo: doing nothing different and continuing to test at 24-28 weeks. It is free, embedded in guidelines, and carries no procurement or regulatory friction. Any pitch for Veritatem is first a pitch against inertia. Beyond that, the realistic alternatives fall into four groups, compared below on the dimensions a clinical buyer actually weighs.

Alternative	What it is	Strengths	Where Veritatem differs
Status quo (OGTT at 24–28 wks)	Standard guideline-based testing	Free, trusted, no friction, universal	Acts months earlier; flags risk while cheap interventions still have time to work
Published academic models	ML risk models in the literature	Often free, peer-reviewed, accurate in-sample	Not products: no CE mark, no integration, no local validation, no support or audit trail
EHR-native risk scores	Risk flags built into hospital EHRs	Already in the workflow; low marginal cost	Usually generic, rarely explainable, rarely validated locally or cleared as a device for this purpose
Commercial maternal-health AI	Vendor platforms / monitoring suites	Funded, polished, sales reach	Broad rather than GDM-early-window specific; explainability and local validation vary

The competitive strategy is not to win on raw model accuracy, because that ground is too easily contested, but to win on the things a published model or a generic EHR flag cannot easily claim: regulatory clearance, locally accepted validation, the early window, explainability, and the support and evidence that come with a real product.

2.5 Product lifecycle

Veritatem's product lifecycle would be longer and more demanding than that of ordinary software because it is a regulated clinical AI product. Before the first sale, the company would need to secure clinical data, develop the model, validate it in healthcare settings and complete conformity assessment. Growth would also depend on clinical evidence and reimbursement approval, not only on marketing. Even after launch, the product would require post-market surveillance, periodic revalidation and monitoring for model drift to ensure that it remains safe and accurate over time.

2.6 Product visualization

Figure 1 shows an illustrative clinician dashboard. Its design intent is that the interface is itself a safety surface. It moves left to right through the clinical reasoning: who the patient is, what the risk tier and calibrated probability are (shown as a tier with an uncertainty band rather than a single false-precision number), why the model produced that score (the per-patient explanation), and what is recommended (a guideline-linked action the clinician can accept or override). Along the bottom runs the audit log that records the model version, the inputs, the output, the human-oversight checkpoint, and the action taken (the raw material for both regulatory traceability and post-market monitoring). For a more hands-on impression, the dashboard can also be explored interactively via the following [link](#).

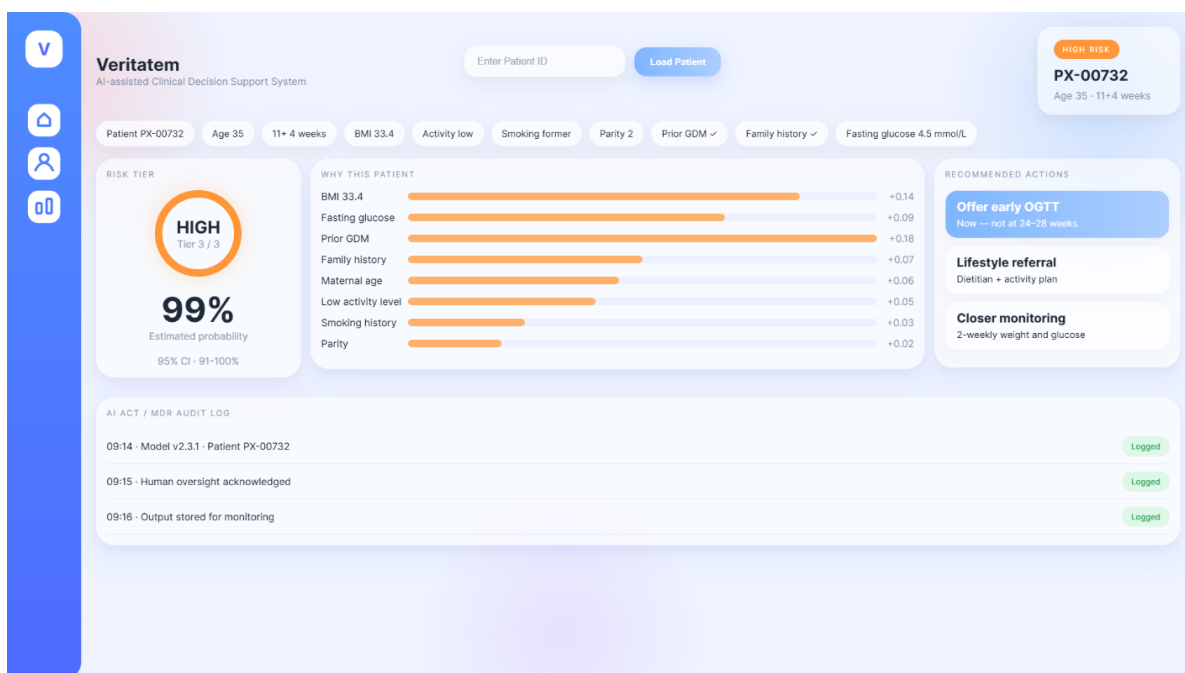


Figure 1. Illustrative clinician dashboard (fictional values)

3. Market Analysis & Strategy

3.1 Market need and size

The denominator is pregnancies. In 2024 the European Union recorded roughly 3.55 million live births (Eurostat). Applying a European GDM prevalence of approximately eleven percent implies on the order of 390,000 GDM cases per year across the EU. But the addressable activity for a first-trimester risk screen is not the GDM cases alone, It is the screening events, because the tool is applied to pregnancies generally to find the at-risk subset. The relevant volume is therefore closer to the full birth cohort that passes through structured antenatal care.

The economic weight of the problem is large. Short-term excess healthcare cost attributable to a GDM pregnancy has been estimated at around €800 per case in an Italian costing study (counting only near-term medical costs), and considerably higher in US analyses (several thousand dollars per pregnancy once obstetric and neonatal complications are included). Multiplying even the conservative European figure by the EU case count puts the near-term cost of GDM in the hundreds of millions of euros annually, before counting the long-term burden of progression to type 2 diabetes.

3.2 TAM, SAM and SOM

The market opportunity is estimated from the number of pregnancies that could be screened and an assumed price of €10 per screening. At the broadest level, the TAM includes all EU pregnancies passing through antenatal care, or around 3.55 million screenings per year, giving an indicative annual value of about €35 million. The SAM is narrower and includes only EU markets where antenatal care is structured, EHR integration is feasible and reimbursement is realistic. This is estimated at around 1.5 million screenings per year, or about €15 million annually.

The initial SOM is much smaller. Austria and Slovenia together represent around 95,000 births per year, and realistic penetration of 15–30% would mean only around 14,000-29,000 captured screenings per year. At €10 per screening, this equals approximately €140,000-€290,000 in annual revenue at maturity. This shows that the beachhead is useful for validation and market entry, but not enough to build a large or profitable company on its own. The economic case therefore depends on expanding into larger EU markets and, later, securing payer or outcomes-based contracts that capture more of the value from avoided complications.

Layer	Definition & method	Volume (approx.)	Indicative annual value
TAM	All EU pregnancies passing through antenatal care that could in principle be screened (EU live births).	~3.55 million screenings / yr	~€35 million / yr at €10/screen

Layer	Definition & method	Volume (approx.)	Indicative annual value
SAM	EU markets with structured antenatal care, feasible EHR integration, and a realistic reimbursement path — a subset of ~5–6 countries.	~1.5 million screenings / yr (assumption)	~€15 million / yr at €10/screen
SOM	Austria–Slovenia beachhead, with realistic penetration over ~5 years at 15–30% of beachhead volume.	~95,000 births/yr beachhead → ~14,000–29,000 screenings/yr captured	~€140k–€290k / yr at maturity

3.3 SWOT analysis

Strengths • Early-window positioning the status quo cannot match • Explainability and audit trail built in as safety features • Cheap inference → favourable marginal economics at scale • Tight, defensible clinical focus (one condition, done well)	Weaknesses • No secured data and no proprietary dataset (the core gap) • Long, expensive pre-market regulatory path before revenue • The model alone is not defensible — it is reproducible • Small founding-stage beachhead volume
Opportunities • Rising GDM prevalence and growing interest in prevention • Emerging RCT evidence that early action can be cost-saving • EHR marketplaces lowering the cost of distribution • Payer appetite for measurable long-term cost avoidance	Threats • Trials could fail to show early information changes outcomes • An EHR vendor could build a “good enough” native flag • Regulatory timelines and rules are shifting (AI Act, Rule 11) • Reimbursement may lag adoption by years

3.4 Target market and positioning

Veritatem targets a narrow but clearly defined niche: first-trimester gestational diabetes risk prediction in structured European antenatal care systems. The initial focus would be Austria and Slovenia, because these markets provide an organised healthcare setting in which the product can be tested, validated and adapted to EU regulatory requirements. This beachhead is not large enough to build the whole business, but it offers a realistic starting point for proving the model, the clinical workflow and the value of early risk detection.

The market is competitive rather than monopolistic. Veritatem would not be replacing one dominant company, but competing with several alternatives, including standard OGTT testing at 24–28 weeks, academic prediction models, EHR-based risk scores and broader maternal-health AI tools.

3.5 Monetization and pricing

Veritatem is a product-based AI offering delivered through a SaaS / AI-as-a-service model. The first revenue stream would be provider subscriptions, where maternity units or antenatal clinics pay recurring fees to access the tool within their clinical workflow. A second option is a pay-per-screening fee, which lowers the entry barrier for smaller clinics and links payment directly to use.

In the longer term, Veritatem could move toward payer contracts, where public health systems or insurers pay per covered pregnancy or based on demonstrated risk reduction. However, this model should come later because it requires strong real-world evidence that the tool improves outcomes or reduces costs. There is also attribution risk, since better pregnancy outcomes depend not only on the algorithm but also on clinicians, patient behaviour and the availability of follow-up care.

A further revenue option is licensing the validated model to EHR vendors, allowing Veritatem to enter existing hospital software systems instead of competing against them directly. Overall, the most realistic strategy is to begin with provider SaaS and pay-per-screening fees, then use clinical evidence from early deployments to support payer contracts and EHR licensing at a later stage.

3.6 Channels and go-to-market strategy

Veritatem's go-to-market strategy would be relationship-led and evidence-based. Because the product is used in a clinical setting, adoption would depend less on advertising and more on trust, validation and professional credibility. The first step would be pilot projects with maternity units or antenatal clinics. These pilots would generate local performance data, show whether the tool fits real clinical workflows and create reference customers for further expansion.

The main channels would include clinical champions, EHR marketplaces, conferences and publications, and later payer or health-technology-assessment dossiers. Clinical champions are important because respected doctors can influence adoption inside hospitals more effectively than direct marketing. EHR marketplaces could reduce integration barriers by placing the tool inside systems clinics already use. Conferences and peer-reviewed publications would help build credibility, while payer dossiers would support the later case for reimbursement and wider funding.

Customer relationships would be high-touch at the beginning. Early users would need onboarding, AI-literacy training, performance reporting and a feedback loop with the clinical team. This is necessary because a regulated clinical AI product cannot be treated as simple self-service software. The strategy should therefore start with providers, who can use the tool directly, and move to payers later once enough real-world evidence exists to justify broader funding.

4. Operations

4.1 AI Technology and Data Requirements

AI Technology and Data Requirements form one section required to develop, deploy, and maintain the product to ensure its successful operation. Veritatem processes structured tabular data. Therefore, our computational needs remain low. Our technical infrastructure uses standard cloud resources for training and hosting. Clinicians access the tool directly inside their existing Electronic Health Record (EHR) workflow via an integration layer. This eliminates the need for a separate login.

In addition, a secure data pipeline is required to collect, validate, and process clinical data while ensuring patient privacy through de-identification and pseudonymization under strict data governance such as the GDPR. Since Veritatem operates in a regulated healthcare environment, particular emphasis is placed on data security, access control, audit logging, and the traceability of model outputs. Therefore, the successful deployment of the system depends not only on the performance of the underlying model, but also on compliance with regulatory requirements and its integration into existing clinical workflows.

4.2 Operations and Key Activities

Key activities include the development, refinement, and continuous improvement of the AI model, as well as the acquisition of clinical data. Ensuring data quality through data cleaning and curation is also a core task. Access to high-quality clinical data cannot be considered a fixed asset and must instead be secured through partnerships with universities and healthcare institutions. That means the venture depends heavily on establishing partnerships that provide access to consent-based local data.

Essential activities include clinical validation, regulatory compliance, and the associated quality management. Market introduction and clinical adoption are equally important to ensure the successful integration of the product into existing workflows. The model must be continuously monitored to detect potential performance drift and, if necessary, be revalidated over time.

4.3 Key Resources and Partnerships

As the business relies heavily on external stakeholders, strategic partnerships play a key role in the successful development and operation of the business. In particular, access to data, regulatory processes, and market adoption depend on establishing and maintaining long-term collaborations with external partners.

- Universities, clinics, and registries provide access to consent-based training and validation data and contribute to the clinical credibility of the product. Since these partnerships have not yet been

established, their successful acquisition represents an important prerequisite for the development of the project.

- EHR vendors represent important integration partners and distribution channels, enabling the seamless incorporation of the software into existing clinical workflows. Maintaining strong relationships with these partners is essential to ensure interoperability and support market adoption.
- A notified body serves as the conformity assessment partner required to obtain CE marking for the device.
- Cloud and infrastructure providers support compliant hosting by ensuring data residency and appropriate security measures.
- Payers and Health Technology Assessment (HTA) bodies are important stakeholders for future reimbursement pathways and for generating the evidence required to support them.

4.4 Compliance and regulatory

Veritatem is legally bound by the Medical Devices Regulation (MDR), the AI Act and the General Data Protection Regulation (GDPR) of the European Union. It also complies with international quality and standardization standards such as ISO and IEC. As especially the AI Act, MDR software classification continue to evolve, the final classification and the associated obligations would need to be confirmed at the relevant point in time.

Within Rule 11 of the Medical Device Regulation (MDR) the EU specifies how medical software is classified based on its risk. Software that provides information for diagnostic or therapeutic decisions is generally assigned to at least Class IIa. Depending on the potential consequences of erroneous results, the classification may be higher. If errors could lead to a serious deterioration in health or the need for surgery, classification as Class IIb is considered. Erroneous decisions resulting in irreversible harm to health or death are classified as Class III. Since Veritatem only supports, and does not replace, medical decision-making, classification as Class IIa appears most likely. However, the final classification would need to be agreed upon with regulatory experts and the notified body.

Pursuant to Article 6(1) in conjunction with Annex I of the AI Act by the EU, an AI system that is itself a medical device, or a safety component thereof, and that is subject to third-party conformity assessment is classified as a high-risk AI system. For Veritatem, this entails several obligations under Articles 9–15 of the AI Act, including the establishment of a risk management system, compliance with data and data governance requirements, the preparation of technical documentation, transparency towards users, human oversight, automatic logging, and requirements concerning accuracy, robustness and cybersecurity. In addition, providers of high-risk AI systems are subject to post-market monitoring obligations under Articles 72 et seq. of the AI Act. The implementation timeline for some of these requirements may change as a result of the 2026 Digital Omnibus process. Despite these possible delays, Veritatem should be developed in compliance with both the AI Act and the MDR, as notified bodies are already incorporating AI-related considerations into medical device assessments.

Article 9 (1) of the General Data Protection Regulation (GDPR) defines processing of health and medical data generally as prohibited. Article 9(2)(a) and (h) provide the necessary legal grounds for Veritatem. These clauses allow the processing of sensitive medical data based on explicit consent or as “necessary for [...], medical diagnosis”. To ensure compliance, we implement strict data minimization, robust access controls, and a comprehensive Data Protection Impact Assessment (DPIA). Architecturally, this points toward innovative privacy-preserving options. This could be pseudonymisation, on-premise or in-region processing, and potentially federated training that lets the model learn from data held at multiple sites without that data leaving them. These choices also help with the data-access problem, because a partner is far more willing to collaborate if its patients' data never has to leave its control.

High quality and standards are essential for medical device manufacturer. The company must operate an ISO 13485 quality-management system, apply ISO 14971 risk management, and follow IEC 62304 for the software lifecycle. These are not optional best practices; they are the framework within which everything else is documented and audited.

A later United States entry would be a separate regime. The FDA would regulate the tool as software as a medical device, most likely through a 510(k) or De Novo pathway, with a Predetermined Change Control Plan to govern model updates, and HIPAA governing the data. This is flagged as a Phase-2 consideration, not part of the EU-first plan, precisely because each jurisdiction is its own cost and timeline.

4.5 Management Team

The success of Veritatem depends on a team that combines expertise in Technology & AI Development, Clinical, Regulatory & Compliance, and Commercial & Strategic Partnerships. Figure 2 shows an organizational chart. The chart illustrates the core capabilities required across the different stages of the project and shows how these functions are complemented by additional operational and administrative roles as the company grows.

1. Technology & AI Development is essential during the initial development and launch phase. Clinical data scientists and machine learning engineers are responsible for the development, maintenance, and continuous improvement of the AI models and the underlying software infrastructure. Experience in gestational diabetes and medical AI is considered essential.
2. Clinical, Regulatory & Compliance functions become increasingly important during the pre-market phase. Regulatory and clinical affairs specialists are responsible for quality management, conformity assessment procedures, compliance with the AI Act and the GDPR, and post-market surveillance activities. In addition, clinical advisors and healthcare professionals provide medical expertise and support the integration of the product into existing clinical workflows while ensuring that ethical considerations are adequately addressed.

- Commercial & Strategic Partnerships gain importance during the growth phase. Responsibilities include market introduction, cooperation with EHR vendors, healthcare institutions, and payers, as well as the development of reimbursement pathways and long-term strategic collaborations.

Initially, the company is expected to operate with a small team with clearly defined responsibilities. As the venture develops, additional personnel and resources will be required, particularly in commercial functions and market development. Salaries are assumed to be aligned with industry standards and constitute a significant component of the overall cost structure.

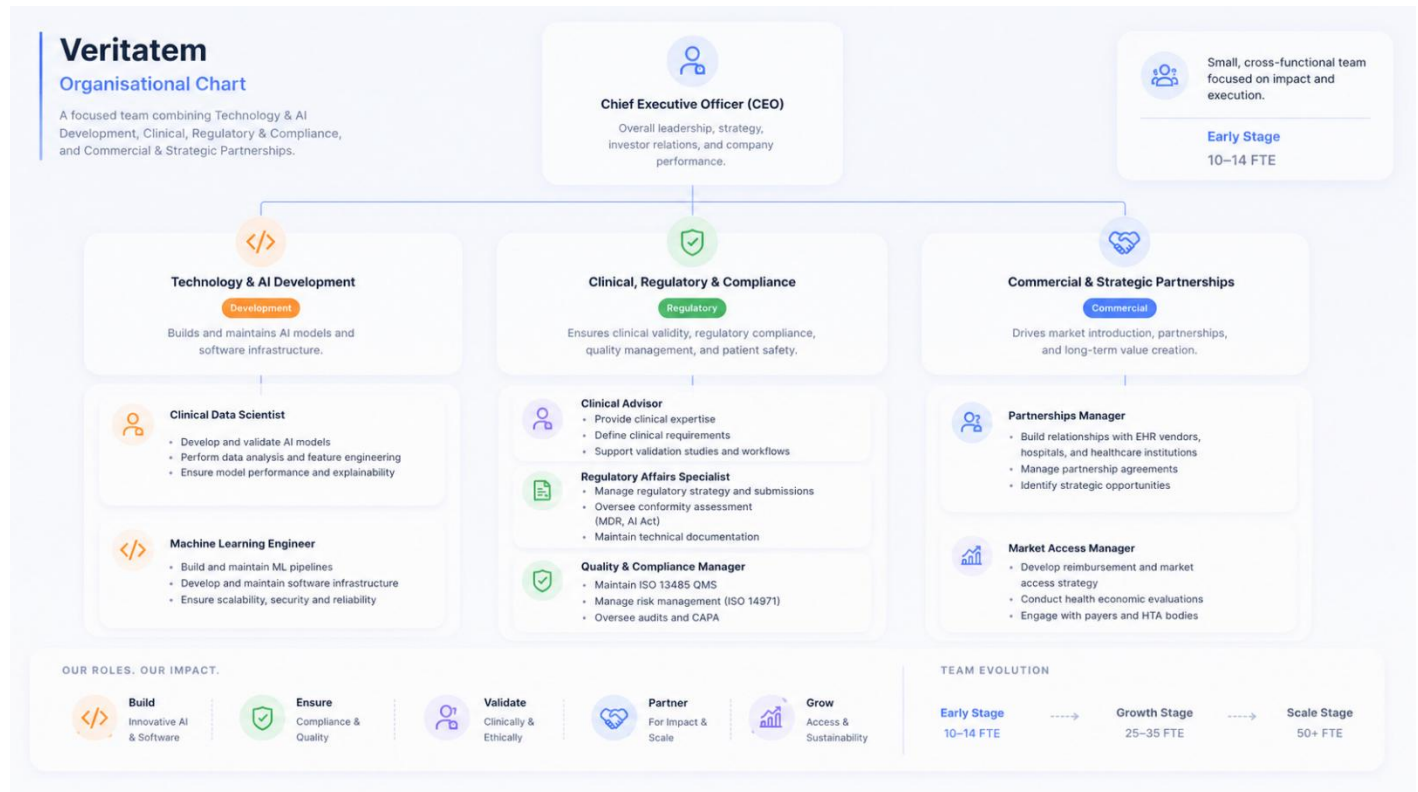


Figure 2. Illustrative organisational chart of Veritatem. The chart presents the core functions and responsibilities required for the venture; individual positions and personnel are to be established as the company develops.

5. Financial Data and Projections

5.1 Monetization recap

Revenue is intended to come from provider SaaS subscriptions and pay-per-screening fees first, payer outcomes-based contracts second, and EHR-vendor IP licensing third (Section 3.5). The conservative core of the model relies on the first two, which do not depend on outcome attribution.

5.2 Cost structure

The dominant cost is personnel costs including clinical data scientists, ML engineers, and especially regulatory and clinical-affairs staff. Those are largely fixed. Major costs also arise from clinical validation studies, conformity assessment activities involving the notified body, quality management, technical documentation, and ongoing post-market surveillance.

In contrast to these fixed costs, cloud and inference costs are variable but comparatively low due to the modest computational requirements associated with structured tabular data. In contrast, the development and maintenance of EHR integrations represent recurring engineering costs. Overall, the cost structure is characterized by high fixed costs and relatively low marginal costs per screening.

5.3 Scale-up cost analysis: 1,000 → 10,000 → 1,000,000 screenings

The table models an illustrative fixed-cost base (assumed at roughly €2.5 million per year for a small regulated-AI team plus ongoing validation and compliance) against a tiny marginal cost per screening (assumed at ~€0.50, covering inference, support, and integration upkeep).

Annual screening volume	Marginal cost / screening	Fixed cost absorbed / screening	All-in cost / screening	Verdict vs ~€10 price
1,000	~€0.50	~€2,500	~€2,500	Deeply negative
10,000	~€0.50	~€250	~€250	Still negative, improving
1,000,000	~€0.50	~€2.50	~€3.00	Strong gross margin

The table illustrates how the cost structure develops as screening volume increases. At low screening volumes, the high fixed costs result in negative unit economics. As the number of screenings grows, fixed costs are distributed across a larger volume, reducing the average cost per screening. Based on the assumptions used in this scenario, the break-even point would be reached at approximately 263,000 screenings. Although this represents a relatively high volume, the low variable costs per screening allow profitability to improve considerably as scale increases.

Post-market surveillance, periodic revalidation, model-drift monitoring, and EHR integration maintenance remain necessary regardless of screening volume and are expected to increase gradually as the installed base expands. New market expanding conducts in high regulatory costs. Reaching one million screenings a year is impossible within the Austria–Slovenia beachhead (which has under 100,000 pregnancies); it requires many countries. Each additional country may introduce further regulatory requirements, language and clinical workflow localization, country-specific validation requirements, and separate reimbursement procedures. Consequently, the average cost per screening is not expected to decrease continuously. Although increasing screening volumes reduce average costs, expansion into additional markets may lead to temporary increases in overall costs. Therefore, the one-million-screening scenario should be interpreted as a mature multi-market stage rather than an immediate expansion scenario.

5.4 Startup costs

The course's reference figure for a generic small business (on the order of a few thousand euros for licenses, deposits, and basic equipment) is the wrong order of magnitude for a regulated clinical AI venture. Based on the Business plan a conservative range to reach a first CE-marked is on the order of €2–5 million. This estimate includes validated deployment, covering a small senior team for roughly two to three years, the initial clinical-validation work, the quality system, and notified-body conformity. The true figure depends heavily on how data access is secured (the availability of academic partnerships could substantially reduce the validation cost) and on the final regulatory classification.

6. Appendix

6.1 Appendix A - Business Model Canvas

The nine-block AI Business Model Canvas below (after Metelskaia et al., 2018) is the analytical engine of this plan. The eleven sections above are where it is argued through. It is reproduced here in full so the mandated template is visible at a glance.

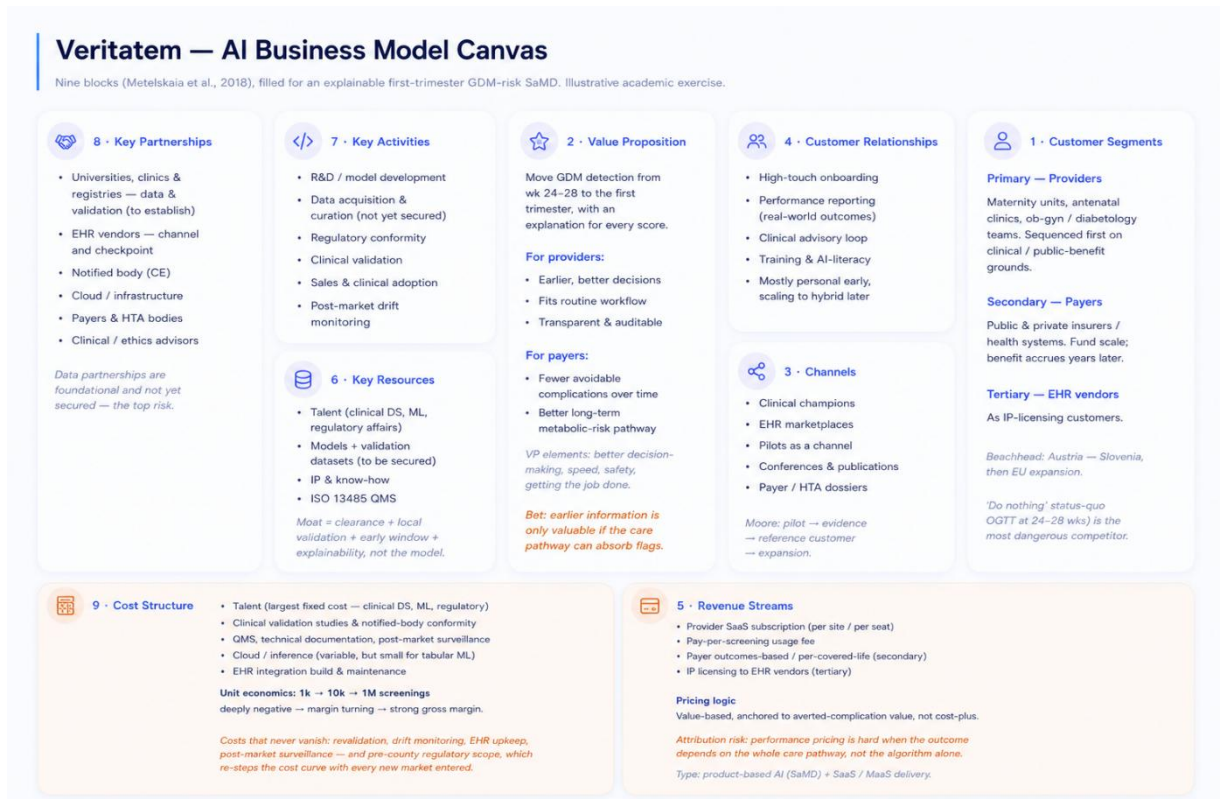


Figure 3. Veritatem on the nine-block AI Business Model Canvas. Items marked “to be secured / to be established” are dependencies the venture does not yet hold — most importantly the local data and the data partnerships.

6.2 Appendix C - Glossary of technical terms

- **GDM (gestational diabetes mellitus):** hyperglycaemia first recognised during pregnancy.
- **OGTT (oral glucose-tolerance test):** the standard diagnostic test for GDM, normally at 24–28 weeks.
- **SaMD (software as a medical device):** software intended for a medical purpose that is itself a regulated medical device.
- **EHR (electronic health records):** digital versions of patients' medical records used for clinical documentation and analysis.
- **Gradient boosting (e.g. XGBoost):** a machine-learning method that combines many decision trees; well suited to tabular clinical data.
- **SHAP-style explanation:** a method that attributes a prediction to the contribution of each input feature, enabling per-patient explanations.
- **Calibration:** how well predicted probabilities match observed frequencies — a model that says “70%” should be right about 70% of the time.
- **Model drift:** degradation of a model's performance over time as the real-world population shifts away from the training data.
- **TAM / SAM / SOM:** total addressable, serviceable available, and serviceable obtainable market.
- **HTA (health technology assessment):** structured evaluation used by payers to decide on coverage and reimbursement.

6.3 Appendix D - References and sources

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21. Regulation (EU) 2016/679 (General Data Protection Regulation, GDPR), especially Article 9 concerning the processing of special categories of personal data.
22. ISO 13485 (quality management systems), ISO 14971 (risk management), IEC 62304 (medical-device software lifecycle).